PROGRESS REPORT

US Green Energy Transition - Visualizing the Past, Present and Future

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ABSTRACT

We analyze the **US shift towards green energy from 2010 through 2030** based on datasets by the US Energy Information Administration (EIA's forms 860 and 923) and the Environmental Protection Agency (EPA's EGRID). The data contains detailed information about the c30K energy generators in the US, such as fuel types, energy generation and capacity, as well as CO2, NOx and SOx emissions, among others. **We analyze the change in the energy generation mix** by fuel type, the trajectory of emissions, cluster states by transition profiles, model bottlenecks and forecast these through 2030 under various scenarios. **We also visualize our analysis via an interactive dashboard** which will allow for user inputs.

REVIEW OF EXISTING SOLUTIONS

In terms of existing energy analytical methodologies, Wang et al [9] assign approaches to four main groups: preprocessing (e.g., fuzzy sets, k-means cluster), deep learning (e.g., DBN, GAN), optimization (e.g., external, trial-and-error) and regression (e.g., linear, logistic; SVM, FNN, LSTM). Kim et al [5] apply time series clustering to smart grid data while Al-Lawati et al [6] apply stochastic optimization to EIA data. Bouziane et al [8] apply LTSM and Monte Carlo simulations to predict energy generation and model uncertainty. An important takeaway is the use of clustering, with Sim et al [7] concluding that it meaningfully improves forecasting accuracy. Mustafa et al [10] review 145 published articles on the use of time series to forecast energy data and conclude that classical approaches (ARIMA, exponential smoothing) are best suited for "linear or regularly seasonal situations", with gray, fuzzy and machine learning (e.g., XGBoost) models better suited for other situations.

In terms of existing visualization approaches, the EIA's Electricity Data Browser has basic charting capabilities while the EPA's eGRID does not have visualization tools. CarbonBrief has state-of-the-art infographics and charts on US electricity generation but no forecasting capabilities. Chen et al [1] provide a review of various screen-, computer- and app-based visualization methods for energy data: maps (dot-based, line-based and area-based GIS maps), 3D models, animation and VR/AR. Cook et al [3] discuss a power flow model utilizing the EIA-860 dataset to visualize crucial information about US electricity generation in a series of line charts and maps. Kunkolienkar et al [4] develop an interactive dashboard which is used to visualize days with similar weather and the corresponding renewable energy power generation on those days.

In general, existing approaches to analyzing and visualizing energy data suffer from at least one of the following three challenges: 1) visualizations are query-driven (not enough of a narrative); 2) the analytical layer is simplistic and provides little informational value beyond the

obvious; and 3) the analysis is backward-looking. We agree with Molnar & al who argue in [2] that the complexity of energy data makes it hard to communicate effectively to stakeholders.

GOALS AND INNOVATIONS

Our project presents a novel combination of a narrative-driven, interactive visualization with a non-trivial analytical layer. The analytical layer models the change in the energy mix over time, gaps / bottlenecks in generator retirements and additions, and energy transition clusters at the generator / plant and state levels - while incorporating the importance of politics and green policies as an additional feature [14], [15], [16], [17]. Importantly, our analytical layer is forward-looking to 2030. **We continue to anticipate being successful** because our novel approach addresses the "complexity of energy data" issue, the quality of the data is good, and collection does not require web scraping / APIs. The over-arching goal of our project is to improve decision making for **the stakeholders that care: corporations, NGOs, asset managers / owners, real estate / operators, governments (federal, state, county) and individuals.** Bonatto et al [18] discuss an application of big data analytics for energy sustainability for the Brazilian public sector while Stecyk et al [19] examine how real-time data and visualization tools can help energy planners interpret complex information quickly and efficiently. Our goals are very much the same.

The successful conclusion of our project should add significant value to stakeholders: the ability to synthesize complex energy data into a coherent narrative and actionable insights; interactive visualizations; and robust forecasting. Our work should help identify green-energy champion vs laggard states (relative to their policies and stated goals), uncover potential bottlenecks and retirement gaps and allow users to model custom scenarios for the future - which is valuable given the discontinuous nature of green energy policies. The results of our project should be able to inform policy and capital allocation at the local and state level.

ANALYTICAL IMPLEMENTATION

We are innovating analytically on three points: 1) forecasting emissions using non-linear regression. 2) gap analysis framework comparing the retirement and additions of power plants.3) state clustering using multi-feature trajectories and policy indicators.

Our analysis decomposes emissions into four multiplicative drivers:

Emissions = Max capacity x Utilization x Heat Rate x Emissions Rate =

$$emissions = \text{max capacity} \cdot \frac{electricity\ generation}{\text{max capacity}} \cdot \frac{fuel\ inputs}{electricity\ generation} \cdot \frac{emissions}{fuel\ inputs}$$

We use the historical time series of these drivers to fit an Ordinary Least Squares or General Additive Model to each of them. Then, we use the models to predict the drivers over the 2025-30 timeframe. The forecasts for the four drivers are multiplied to arrive at emissions forecasts for 2025-30. The use of non-linear OLS or GAM captures any non-linear behaviors while maintaining

interpretability for stakeholders, which is a key goal for our project. We are implementing the models using python libraries (pandas, pyGam, statsmodels, scikit0-learn).

We are currently building out our gap analysis framework., Our approach consists in extrapolating the trend in retirements and additions as % of existing capacity. Users will be able to vary the speed of retirements and additions in order to see how sensitive energy capacity gaps are to a different set of assumptions.

In terms of clustering, we are experimenting with different combinations of features, different values for the 'k' parameter in k-means and 'epsilon/minimum number of members' for DBSCAN. We may use other clustering techniques as well.

VISUALIZATION IMPLEMENTATION

The goal is to have an integrated and interactive data visualization that tells a coherent story. The backdrop of our visualization consists of a map of the USA where a metric and a state can be selected - the remaining states are colored according to the value of that metric. Clusters can also be chosen as a visualization variable. A slider can be used to display the evolution of the metric and clusters over time. Below the map, a user can compare the selected state with up to four additional states on up to nine different metrics. The nine charts are line-based and show in more detail the evolution of nine metrics over time, comparing the selected states with the other four states as well as the US average.

As of now, we have extracted, cleaned and wrangled the raw data and incorporated it into our preliminary dashboard. We need to finalize the calculation of some of the metrics which has proven challenging due to different granularity of the raw data and the biennial nature of the emissions data (compared to the annual disclosure for forms 860/923). We have also started to work on visualizing the planned retirement and addition of power plants which drives our capacity gap analysis. Related to that, we are putting in place the option for users to set their own assumptions for the pace of retirements, additions, utilization and emissions intensity. We believe that the consolidation of the various visualizations into one dashboard will offer a novel narrative-driven approach. This narrative driven approach is key for us as our main goal is to advise stakeholders and policy makers.

EVALUATION

We plan to start evaluation of results as planned, in the first week of April.

Evaluation of our analytical results will consist of: hindcast and stability tests for the gap analysis and forecasts, and silhouette scores and domain validation for clustering, Hindcast tests will involve comparing predicted values with actual outcomes (e.g., for 2015-24) while stability tests will help us understand whether our forecasting model is sensitive enough (or too sensitive) based on the magnitude of the estimated coefficients. Silhouette scores are the accepted way of measuring distinctiveness in clustering while domain validation is the common sense test for "Can we label each cluster in a way that conforms to reality?"

Evaluation of our visualization will be based on: a task-based usability evaluation (5-7 users, quantitative metrics), a comparative evaluation (e.g., vs EIA Electricity Data Browser), and a heuristic evaluation (2-3 users, Nielsen's usability heuristics). Our review of [11], [12] and [13] shows that ground-truth data, systematic benchmarking and statistical measures (prediction error, bias) are also well-established. Regarding the usability testing, we plan to set queries such as: "Find the state with the highest percent change in gas-powered energy generation for the year 2016". How easy was it to find that information? For a control we would prompt with the same questions using a comparative analytical browser for energy in the US such as the EIA data browser. In addition, we would request subjective feedback from the users in terms of their overall impressions of the visualizations and user interface. Ideally, we wouldn't just use peers from OMSA / OMSCS but have a representative sample of the general population. The visualizations should be intuitive enough that most people can use them regardless of their backgrounds.

Previous Gantt chart

Activity	Assigned to	Start	Duration(days)
Project scoping, prep, lit review	All	9-Feb	22
Proposal doc, slides, video (due Mar 6th)	All	2-Mar	5
Progress report (due Apr 3rd)	All	27-Mar	8
Final report, poster video (due Apr 24th)	All	17-Apr	8
Data Collection, Wrangling (Python etc)	Stefan/Krris/Ayo	7-Mar	14
Data engineering, databases (SQLite etc)	Ayo/David/Dylan	7-Mar	14
Analysis, big data (clustering, Hadoop etc)	Stefan/Krris/Aaron	14-Mar	14
Visualization (D3 etc)	All	21-Mar	21
App development (Flask, Streamlit etc)	Ayo/David/Dylan	28-Mar	21
Testing & refining	All	4-Apr	14

Current Gantt chart

Activity	Assigned to	Start	Duration(days)	End Date	Status	Revised End Date
Project scoping, prep, lit review	All	9-Feb	22	3-Mar	Done	N/A
Proposal doc, slides, video (due Mar 6th)	All	2-Mar	5	7-Mar	Done	N/A
Progress report (due Apr 3rd)	All	27-Mar	8	4-Apr	Done	N/A
Final report, poster video (due Apr 24th)	All	17-Apr	8	25-Apr	Not started	N/A
Data Collection, Wrangling (Python etc)	Stefan/Krris/Ayo	7-Mar	14	21-Mar	Done	N/A
~~Data engineering, databases (SQLite etc)~~	~~Ayo/David/Dylan~~	~~7-Mar~~	~~14~~	~~21-Mar~~	~~Removed~~	~~N/A~~
Analysis (Pandas dataframes)	Stefan/Krris/Aaron	14-Mar	14	28-Mar	In progress	11-Apr
Visualization (D3 etc)	All	21-Mar	21	11-Apr	In progress	N/A
App development (Flask, Streamlit etc)	Ayo/David/Dylan	28-Mar	21	18-Apr	In progress	N/A
Testing & refining	All	4-Apr	14	18-Apr	Not started	N/A

All team members have contributed equally

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